

Author Contributions Statement

IEEE Transactions on Power Delivery

“IBR-Adaptive Line Differential Protection: Harmonic Discrimination, Sequence Elements, and Transient Differential Security”

The contributions of each author are described below using the CRediT (Contributor Roles Taxonomy) framework.

CRediT Role	Anoop V. Eluvathingal	K. Shanti Swarup
Conceptualisation	Originated the five-element 87L chain concept; identified TDCS-87L as the primary detection mechanism for symmetric IBR faults	Provided the strategic research framing; guided scope and novelty positioning relative to prior art
Methodology	Derived the physics-constrained adaptive slope $k_m^*(k_{ibr})$; developed the 87LN $\rho_2 = 2$ proof; designed the DDR model selector; formulated the TDCS security bound and its formal proof	Reviewed and validated all analytical derivations; confirmed alignment with IEC 60255 and IEEE C37.243 relay standards
Software	Implemented the full protection chain in Python/ANDES; coded the Monte Carlo engine (20 000 trials, four groups); developed the deterministic test harness (30 scenarios)	—
Validation	Ran all simulations; performed Wilson CI analysis; verified zero-regression on SG reference cases	Independently verified the 15.5 % TDCS security margin computation and the $\kappa_n = 1$ condition number proof
Formal Analysis	Proved Propositions 1–3 (87LN sensitivity, TDCS detection floor, PV unit condition number)	Reviewed all proofs for mathematical rigour and physical interpretation
Investigation	Designed and executed the parametric sensitivity study; identified the $k_{ibr} = 0.12$ pu TDCS/87L transition threshold	Advised on study network parameters and CT class selection
Writing — Original Draft	Wrote the full manuscript	—
Writing — Review & Editing	—	Reviewed and edited all sections; approved the final submission version
Supervision	—	Supervised the overall PhD programme of which this paper forms a part
Funding Acquisition	—	Secured SERB and IIT Madras research funding supporting this work

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Anoop V. Elovathingal: Conceptualisation (lead); Methodology (lead) — derived all analytical results including the adaptive slope formula, the 87LN sensitivity proposition, the TDCS detection floor proof, and the PV unit condition number; Software (sole); Validation (lead); Formal Analysis (lead); Investigation (lead); Writing — original draft (sole).

K. Shanti Swarup: Conceptualisation (supporting) — strategic research framing; Methodology (supporting) — reviewed all derivations and verified standard compliance; Validation (supporting) — independently verified key numerical results; Formal Analysis (supporting) — reviewed all propositions; Writing — review and editing (sole); Supervision; Funding acquisition.